

BARKSDALE AFB, LA, USA

WMO: 722485

Lat: 32.500N

Lon: 93.667W

Elev: 51

StdP: 100.72

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13944

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.3	(c) -2.5	(d) -12.0	(e) 1.3	(f) -1.0	(g) -9.3	(h) 1.7	(i) 1.4	(j) 10.9	(k) 12.5	(l) 9.8	(m) 11.2	(n) 2.7	(o) 330	(p) 0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	37.3	24.2	35.9	24.3	34.7	24.4	26.3	32.4	25.8	31.7	25.5	31.3	3.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.1	20.3	27.4	24.5	19.6	27.4	24.0	19.0	27.4	81.5	32.4	79.7	31.6	78.1	31.4	32.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB
(a) 9.2	(b) 8.2	(c) 7.3	(d) 7.8	(e) 38.4	(f) 2.3	(g) 2.2	(h) -9.5	(i) 40.0	(j) -10.9	(k) 41.3	(l) -12.2	(m) 42.5	(n) -13.9	(o) 44.0	(p) 44.0	
(a) 9.2	(b) 8.2	(c) 7.3	(d) 8.7	(e) 27.3	(f) 2.3	(g) 1.3	(h) -10.3	(i) 28.2	(j) -11.7	(k) 29.0	(l) -13.0	(m) 29.7	(n) -14.6	(o) 30.7	(p) 30.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.8	8.2	10.7	14.4	18.3	22.9	26.9	28.4	28.3	25.1	19.1	13.1	9.6
	DBStd	8.36	5.74	5.71	5.15	4.20	3.36	2.40	2.18	2.48	3.20	4.37	5.02	5.48
	HDD10.0	283	103	56	20	2	0	0	0	0	0	1	25	77
	HDD18.3	1221	316	219	140	52	6	0	0	0	2	46	167	274
	CDD10.0	3494	48	76	157	250	400	506	569	568	453	282	119	66
	CDD18.3	1391	2	6	19	51	148	256	311	309	206	68	11	5
	CDH23.3	14541	7	31	138	411	1316	2718	3627	3575	2015	614	69	18
	CDH26.7	6780	0	2	19	87	475	1289	1865	1892	947	195	10	0
(14) Wind	WSAvg	3.2	3.6	3.7	3.9	4.0	3.4	3.0	2.7	2.5	2.4	2.7	3.1	3.4
(15) Precipitation	PrecAvg	1313	125	107	134	128	112	108	92	77	94	101	110	124
	PrecMax	1896	336	210	376	427	256	334	266	213	203	442	273	206
	PrecMin	734	9	6	23	31	16	10	16	5	7	0	25	9
	PrecStd	293	71	58	76	90	72	75	62	53	55	92	60	47
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.2	26.4	28.6	31.2	33.9	37.4	38.1	39.9	37.2	32.6	28.0	25.3
		MCWB	17.6	18.0	19.5	21.1	23.2	24.1	24.0	24.0	23.0	22.1	20.8	19.6
	2%	DB	22.3	24.0	26.8	29.0	32.3	35.6	36.8	37.6	35.1	30.7	25.2	23.1
		MCWB	16.2	17.6	18.7	20.8	23.0	24.2	24.2	24.5	23.3	21.6	19.0	18.6
	5%	DB	20.0	22.1	24.8	27.5	31.0	34.0	35.3	36.0	33.4	29.0	23.2	21.1
		MCWB	15.5	17.1	17.6	19.7	22.8	24.4	24.6	24.5	23.4	20.5	18.1	18.0
	10%	DB	17.6	20.0	22.7	26.0	29.6	32.6	34.0	34.5	32.0	27.0	21.4	19.1
		MCWB	14.4	15.8	16.9	19.1	22.4	24.4	24.8	24.5	23.0	20.1	17.6	16.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.4	20.4	21.2	23.4	25.4	26.5	26.7	27.1	26.0	24.6	22.2	21.2
		MCDB	22.5	23.2	25.2	27.4	30.6	31.5	32.8	33.7	31.0	29.0	25.4	23.1
	2%	WB	18.0	19.2	20.2	22.3	24.5	25.8	26.1	26.4	25.3	23.8	20.9	19.9
		MCDB	20.8	22.2	24.2	26.4	29.5	31.1	32.5	32.7	30.5	27.9	23.6	22.0
	5%	WB	16.9	18.2	19.3	21.5	23.8	25.4	25.7	25.8	24.8	22.8	19.8	18.8
		MCDB	19.0	21.0	23.1	25.5	28.7	30.9	31.9	31.9	29.8	26.2	22.2	20.9
	10%	WB	15.1	16.9	18.3	20.7	23.1	25.0	25.4	25.5	24.1	21.7	18.2	17.0
		MCDB	16.9	19.3	21.8	24.4	27.9	30.5	31.4	31.4	29.1	24.7	20.7	18.8
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	11.3	11.8	12.1	11.1	10.6	10.6	11.2	12.1	12.8	12.0	11.0
		MCDBR	14.2	13.3	13.3	13.1	12.0	12.0	11.9	13.1	13.5	14.3	13.4	12.8
		MCWBR	9.3	8.2	6.7	6.0	4.3	3.3	2.9	3.5	4.1	6.0	8.1	8.8
	5% WB	MCDBR	10.9	11.0	11.2	10.5	10.3	10.3	10.5	11.0	10.9	10.9	11.4	11.3
		MCWBR	8.6	8.1	6.9	6.0	4.4	3.7	3.4	3.5	3.7	5.9	8.1	8.9
(40) Clear-Sky Solar Irradiance	taub		0.331	0.342	0.366	0.399	0.429	0.463	0.477	0.476	0.422	0.365	0.351	0.335
	taud		2.502	2.470	2.410	2.318	2.294	2.240	2.230	2.226	2.346	2.479	2.467	2.494
	Ebn at Noon		890	913	912	891	863	829	816	812	845	876	856	863
	Edh at Noon		87	99	113	129	133	140	141	139	119	97	89	83
(44) All-Sky Solar Radiation	RadAvg		2.70	3.03	4.29	5.34	5.90	6.25	6.25	5.87	5.03	4.06	2.95	2.34
	RadStd		0.26	0.51	0.38	0.34	0.42	0.47	0.47	0.50	0.45	0.56	0.33	0.30

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d) +0.76	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (31 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page